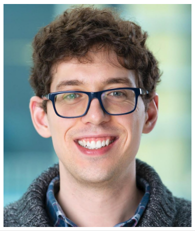

Last Updated on 7th June 2026

Mario Lavanga

Senior Research Engineer | Health AI & Causal Inference
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EDUCATION

KU LEUVEN
PhD in Electrical Engineering
Grad. September 2020 | Leuven, Belgium | Phd Graduation Video

POLITECNICO DI MILANO
MS in Biomedical Engineering
Grad. December 2015 | Milan, Italy | Final Grade: 100/100 cum laude

BS in Biomedical Engineering
Grad. December 2015 | Milan, Italy | Final Grade: 100/100 cum laude

ALTA SCUOLA POLITECNICA
Double degree with Politecnico di Torino
Grad. December 2015 | Turin, Italy

LINKS

- Mario Lavanga
- Mario Lavanga
- Mario Lavanga

SKILLS

PROGRAMMING
Python • Matlab • C# • Bash • Powershell

FRAMEWORKS
Azure • Azure AI Foundry • Azure DevOps • Tensorflow • Git

STATISTICS & MODELLING
Time-Series & Longitudinal Modelling • Causal Inference & Discovery (Bayesian) • Uncertainty Quantification (VAE, SNPE) • Hypothesis Testing

LANGUAGES
Italian | Mother tongue • English, German, French | Fluent • 汉语, Dutch | Basic

AI CODING TOOLS
Claude Code • Codex • Cline • Gemini CLI • MCP integrations

ABOUT ME

I build deep-learning and signal-processing models on physiological data, then prove they hold up on real hospital data before they reach a patient. My methods: time-series modelling, causal inference, supervised learning, and uncertainty quantification, on observational ICU and NICU data under clinical and regulatory constraints. My background spans the aging brain, real-time ICU monitoring, and neonatal stress detection. Offstage I act in amateur theatre and keep collecting languages.

RESEARCH

HAMILTON MEDICAL AG | Project Leader
December 2024 –Present | Domat/Ems, Switzerland
Leading cloud ventilation-management features from research through IP protection and regulatory submission.

HAMILTON MEDICAL AG | Senior Research Engineer
June 2022 –Present | Domat/Ems, Switzerland
Built deep-learning and rule-based models for patient-ventilator asynchrony, respiratory mechanics, and patient effort, and deployed the asynchrony detectors on both the ventilator itself (embedded, real-time) and a cloud platform. Working on patent applications.

TNG - INS | Postdoctoral researcher
Oct 2020 –June 2022 | Marseille, France
Collaborated with Prof. Viktor Jirsa and Prof. Svenja Caspers to investigate virtual brain aging (Digital Twin) using causal inference and uncertainty quantification (Variational Autoencoders) on anatomical data. Part of the Human Brain Project, showcased on EBRAINS (code).

BIOMED GROUP - STADIUS | PhD Fellow
Jan 2016 –Sep 2020 | Leuven, Belgium
Developed a Perinatal Stress Calculator using multimodal biosignals (ECG, HRV, EEG) in collaboration with Prof. Sabine Van Huffel and Prof. Gunnar Naulaers. Focused on data-driven stress quantification in premature babies via longitudinal analysis and robust features in noisy clinical environments. Collaboration with UZ Leuven.

SELECTED PUBLICATIONS & CODE

M. Lavanga et al., The virtual aging brain: Causal inference supports interhemispheric dedifferentiation in healthy aging. NeuroImage , 2023.

M. Lavanga et al., The effect of early procedural pain in preterm infants on the maturation of EEG and heart rate variability. Pain , 2021.

Open ML: In-browser ECG classifier —calibrated, ONNX, on-device; 0.92 macro-AUROC on PhysioNet (PTB-XL). github.com/mlavanga/ecg-ptbxl.

Full list on Google Scholar —612 citations, h-index 13, i10 18.

AWARDS AND VARIA

- Guest Editor for Frontiers in Network Physiology (2021-2022) | Talk at University of Virginia hosted by Prof. Randall Moorman.
- Falling Walls Lab Leuven Participant (2019) | KU Leuven Tech Transfer Laureate.
- FWO Strategic Basic Research Fellowship Recipient (2016).